Nazrul Huda Shanto

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Summary

Software and ML engineer with 3+ years of combined industry and research experience building LLM agents, RAG systems, backend APIs and ML on AWS and Kubernetes. First-author LLM-agent paper under review (IEEE BigData 2026); IEEE TCC paper (29+ citations).

Education

M.S. in Computer Science, Oklahoma State University, Stillwater, OK – GPA: 3.82/4.0 — Jan 2024 – May 2026
B.S. in Computer Science, BRAC University, Dhaka, Bangladesh – CGPA: 3.36/4.0 — May 2018 – May 2022

Technical Skills

AI/ML & LLMs: LangGraph, LangChain, MCP, OpenAI API, RAG, pgvector, LLM evaluation, PyTorch, scikit-learn, Hugging Face
Languages & Backend: Python, SQL, Java, TypeScript; FastAPI, Flask, Django, Spring Boot, Node.js, Next.js, React, JWT/JWKS
Cloud & DevOps: AWS (Lambda, ECR, S3, Textract, MQ), Docker, Kubernetes, Istio/Envoy, Prometheus, GitHub Actions, Linux
Data & Databases: PostgreSQL/PostGIS, Redis, MongoDB, Supabase, Neon, pandas, NumPy, SPSS; schema design, row locking

Experience

Graduate Research Assistant — Jan 2024 – May 2026
Oklahoma State University — *Stillwater, OK*

- Built an LLM agent (LangGraph, FastMCP, Flask, Redis, PostgreSQL, Docker) with 8 custom MCP tools that lets non-experts run and verify zero-knowledge proofs of statistical tests on private clinical data; first-author paper under review at IEEE BigData 2026.
- Designed skeptical tools, 20 server-side design decisions that check, fill in or override LLM tool arguments; raised task completion from 70.2% to 93.0% (Qwen 3 32B), about +23 points on all 3 model families, and cut response time from 16.6 s to 12.4 s.
- Designed the evaluation: an 82-query test suite and a 5-configuration ablation (6,000+ runs); analyzed 6,371 result records, found 198 silently misrouted status checks (fixed to 0), and built a rule-based grader that matched human labels on 150 of 151 replies.
- Contributed new APIs and a verification bug fix, now merged into the open-source CoSMeTIC zkSNARK framework.
- Built per-service DDoS detection for Kubernetes/Istio: six traffic features from Envoy/Prometheus telemetry over 4,705 labeled 20-second windows; per-service classifiers (Random Forest, SVM and others) reached 94.0–97.8% accuracy (ICOIN 2027, under review).
- Deployed the detector as one least-privilege pod with a JSON model registry and Isolation Forest fallback; parallel per-pod inference cut detection time by 55.9% (37.4 vs. 84.8 ms), and traffic reroutes to a backup version automatically.

Junior Software Engineer — Feb 2023 – Dec 2023
Tirzok Private Ltd. — *Dhaka, Bangladesh*

- Led ML document validation for CliQloan, a mortgage loan origination system: classifiers sort and trim documents before AWS Textract extracts key fields; 45× faster than manual checks (scikit-learn, Spring Boot, PostgreSQL, AWS Lambda/ECR).
- Built RESTful APIs and JWT-based role management (Node.js, MongoDB) for Janatar Sarkar, a government–citizen portal serving 80,000+ citizens; optimized aggregation pipelines to cut query latency by 40%.
- Built a Flask and PostgreSQL currency-conversion API (5 currencies) for a payroll system; deployed services on Linux cloud servers.

Undergraduate Research Assistant — Oct 2020 – Jan 2023
BRAC University — *Dhaka, Bangladesh*

- Built a cloud image storage and retrieval framework with a compression module (up to 27% smaller files, 54% less waiting time, validated with SSIM on COCO); corresponding author, IEEE Transactions on Cloud Computing.
- Ran SPSS regression and descriptive analysis on barriers to primary education in remote regions (ACM COMPASS 2026), and led fieldwork for two digital-divide studies (73 participants, 15 villages).

AI Instructor, NSF RET, Oklahoma State – taught Python, ML and Transformers to 10 K–12 teachers — May 2025 – Aug 2025

Projects

FilingCheck – LLM Agent for SEC 10-K Filings | filingcheck.vercel.app | *FastAPI, pgvector, Next.js* — Oct 2026 – Present

- Built retrieval over 75 SEC 10-K reports from 25 large US companies (42,581 passages, pinned by accession number); an output-only data check found 54 problems that 134 passing unit tests missed.
- Measured vector, keyword and hybrid search on 40 questions with company and year filters (vector recall@10 0.80, MRR 0.511); hybrid did not win.
- Shipped a FastAPI API and a Next.js page with citation chips; 498 unit tests in GitHub Actions CI; live on Vercel, Render and Neon.
- In progress: server-side checks on company, fiscal year and metric before each tool call, and XBRL-checked labels on every number.

FwdStar – Freight Marketplace | Technical Lead and Sole Architect | *Next.js, FastAPI, PostGIS, Docker* — Jan 2026 – Present

- Turned a client's verbal needs into a 2,695-line spec and shipped the MVP in nine phases (sign-up, fleet management, load board, awards, GPS tracking, SMS alerts) with a bilingual frontend and 50+ REST endpoints.
- Designed a PostgreSQL/PostGIS schema (20+ tables, 13 migrations) with state machines and SELECT FOR UPDATE locks so a load is never awarded twice; layered security with JWT/JWKS, CSRF, OTP rate limiting and audit logs.

Selected Publications

IEEE TCC 2023 (corresponding author, 29+ citations) · IEEE BigData 2026 (first author, under review) · ICOIN 2027 (under review) · ACM COMPASS 2026 · NSysS 2022